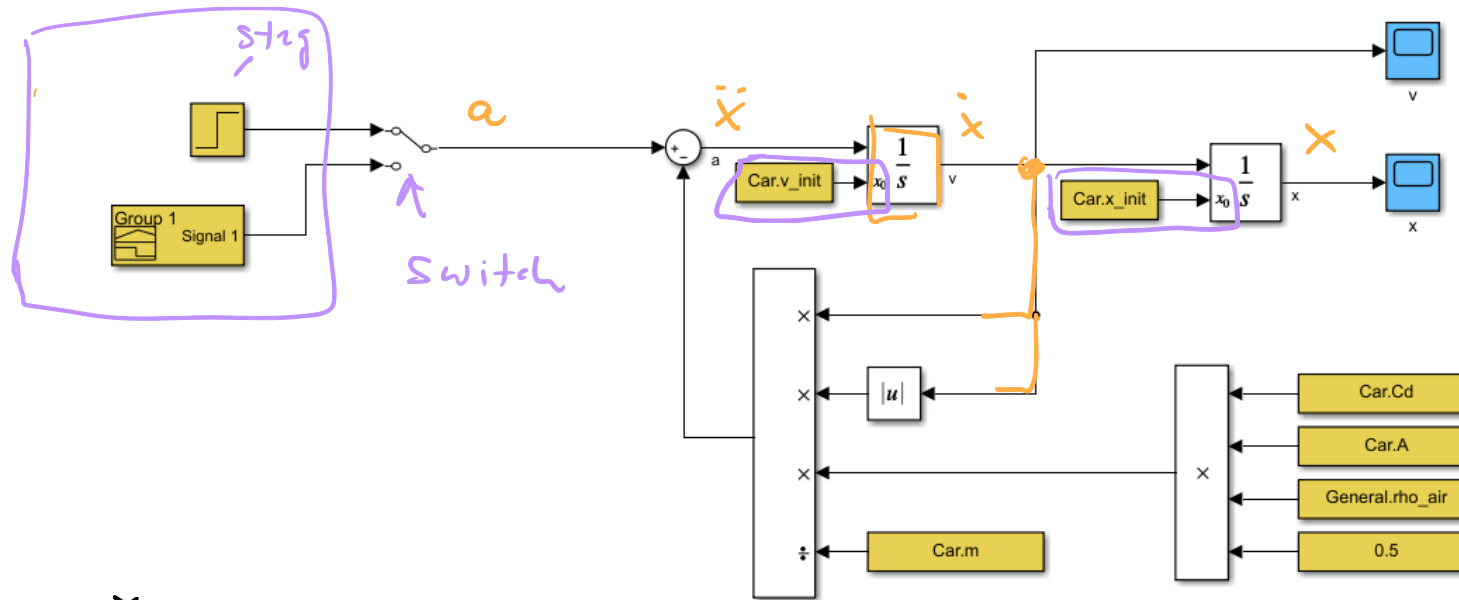


$$F - F_v = m \cdot \ddot{x} \quad (1)$$

$$F_v = \frac{1}{2} v^2 \cdot c_d \cdot A \cdot \delta_{Luft}$$

$$= \frac{1}{2} |v| \cdot v \cdot c_d A \delta_{Luft}$$

$$\stackrel{(1)}{\Rightarrow} \ddot{x} = \underbrace{\frac{F}{m}}_a - \frac{1}{2} \underbrace{|v| \cdot v}_{\dot{x} \cdot |\dot{x}|} \cdot c_d \cdot A \delta_{Luft} \cdot \frac{1}{m}$$

